

Chapter 6 — Exercises

Multiple-Choice Questions

Algorithm chains & pipelines

How to Use These Slides

- **12 multiple-choice** questions on Chapter 6 concepts
- Pick **one** answer (A–D) for each question
- The **answer key** with brief explanations is at the end
- Try to answer before revealing the solutions

Topics: data leakage, the `Pipeline` class, parameter naming, `make_pipeline`, `named_steps`, and grid-searching workflows.

Q1. Scaling **all** training data before cross-validation causes:

- A. data leakage and over-optimistic scores
- B. faster convergence
- C. underfitting
- D. no problem at all

Q2. Any step that learns from data must be fit on:

- A. the whole dataset
- B. the test folds
- C. the training folds only
- D. a random subset of features

Q3. A **Pipeline** chains steps into:

- A. several independent models
- B. a single estimator with `fit` / `predict` / `score`
- C. a plotting utility
- D. a cross-validation splitter

Q4. In a pipeline, every step except the last must implement:

- A. `predict`
- B. `score`
- C. `fit_proba`
- D. `transform`

Q5. To tune the **C** of an SVM step named **svm**, the grid key is:

- A. `svm__C`
- B. `svm.C`
- C. `C__svm`
- D. `svm-C`

Q6. `make_pipeline` names each step:

- A. with numbers only
- B. automatically, using the lowercase class name
- C. after its parameters
- D. randomly

Q7. To inspect a fitted step inside a pipeline, use:

- A. `best_params_`
- B. `steps_only`
- C. `named_steps`
- D. `get_dummies`

Q8. Compared to scaling leakage, feature-selection leakage is:

- A. identical in effect
- B. always harmless
- C. easier to spot
- D. far worse — it can make random data look predictive

Q9. To grid-search **which model to use, you pass:**

- A. a list of parameter-grid dictionaries
- B. a single float
- C. two separate test sets
- D. no `param_grid` at all

Q10. Setting a pipeline step to `None` in a grid means:

- A. delete the pipeline
- B. skip that step
- C. reset all parameters
- D. use the default scaler

Q11. Inside cross-validation, a pipeline refits its scaler:

- A. once on all the data
- B. never
- C. in every fold
- D. only on the test set

Q12. Adding more parameters to a grid search makes the number of fits:

- A. stay the same
- B. decrease
- C. grow linearly
- D. grow exponentially

Answer Key

Solutions & Explanations

Answers 1–6

Q	Ans	Why
1	A	The scaler sees validation folds → leakage
2	C	Learn preprocessing on training folds only
3	B	<code>Pipeline</code> = one estimator (<code>fit</code> / <code>predict</code> / <code>score</code>)
4	D	All but the last step need a <code>transform</code> method
5	A	Use <code>svm__c</code> (step name + <code>__</code> + parameter)
6	B	<code>make_pipeline</code> uses the lowercase class name

Answers 7–12

Q	Ans	Why
7	C	<code>named_steps</code> maps step names → fitted estimators
8	D	Feature-selection leakage is far worse than scaling
9	A	Pass a list of dicts to swap estimators
10	B	<code>None</code> means skip that step (e.g. no scaling)
11	C	The scaler is refit per fold → no leakage
12	D	Combinations multiply → exponential growth

Well done!

Review Chapter 6 for any you missed